Here are the answers to the periodic table crossword puzzle based on the letter counts of the grid:

Across

- **4. What is column 18? NOBLEGASES** (10 letters)
- **6. What element has the symbol S? SULFUR** (6 letters)
- **7. What is another name for rows? PERIODS** (7 letters)
- **9. What kind of element is Helium? GAS** (3 letters)
- **13. The amount of protons is equal to the what? ATOMICNUMBER** (12 letters)
- **15. What are the vertical lines called? COLUMNS** (7 letters)
- **17. All elements have protons neutrons and what? ELECTRONS** (9 letters)
- **18. What is another word for columns? GROUPS** (6 letters)
- **19. Columns 3-12 are what kind of metals? TRANSITION** (10 letters)
- **20. What is column 1? ALKALIMETALS** (12 letters)

Down

- **1. Weighted average of mass of an element? ATOMICWEIGHT** (12 letters)
- **2. What is oxygen? NONMETAL** (8 letters)
- **3. What is column 2? ALKALINEEARTH** (13 letters)
- **5. What element has the atomic mass of 1.008? HYDROGEN** (8 letters)
- **8. Who created the Periodic Table? MENDELEEV** (9 letters)
- **10. AU is what for Gold? SYMBOL** (6 letters)
- **11. What are the horizontal lines called? ROWS** (4 letters)
- **12. How many protons does boron have? FIVE** (4 letters)
- **14. What is column 17? HALOGENS** (8 letters)
- **16. What is hydrogen? NONMETAL** (8 letters)